Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
6.	(a)			There would be zero resistance / a short circuit / resistance would get too low (1) producing a [very] high current / safety resistor reduces the current through the ammeter / overload the ammeter (1) N.B. Treat as neutral reference to voltage		2		2		2
	(b)	(i)		0.400 Accept 0.4		1		1	1	1
		(ii)		10 [Ω]		1		1		1
		(iii)		20 [Ω]		1		1	1	1
		(iv)		Horizontal scale of 0 to 10 in 2 (A^{-1}) per 2 cm square (1) 5 points plotted correctly < 1 small square tolerance (1) 4 or fewer points plotted correctly < 1 small square tolerance award 0 marks Straight line of best fit from the plotted points < 1 small square tolerance (1) N.B. doesn't need to extend to the origin	1	1 1		3	3	3
	(c)			R = 70 [Ω] or their answer from (b)(iii) **ecf** $\times$ 3 + r **ecf** (1) accept 60 [Ω] 5.8 ± 0.2 (value taken from graph) (1) Accept 5 ± 0.2 Current = $\frac{1}{5.8}$ = 0.17 [A] (accept 0.18 - 0.21 [A]) (1)			3	3	3	3

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
	(d)	(i)		Voltage = gradient = $\frac{24}{2}$ or $\frac{48}{4}$ etc (taken from the graph) (1) = 12 [V] (1) Accept correct substitution into $V = IR$ and correct answer		2		2	2	2
		(ii)		Take corresponding values of resistance and mean current from the table (1) Use of $V = IR$ to find V / multiply them (1) Accept use of $P = IV$ for 1 mark Don't accept use a voltmeter	1	1		2		2
				Question 6 total	**2**	**10**	**3**	**15**	**10**	**15**